

Ex.no: 12

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NMAP TO DISCOVER LIVE HOST USING ARP SCAN, ICMP SCAN, AND TCP/UDP PING SCAN IN NMAP.

AIM:

Learn and implement NMAP host discovery technique using ARP, ICMP, TCP, UDP ping to identify live hosts on a network and compare their effectiveness.

Procedure:

- ARP scan: `nmap -sn -PR 10.10.10.0/24`
- ICMP Echo/timestamp/mask: `nmap -sn`
- TCP SYN/ACK (-PS, specific ports)
- UDP: `nmap -sn -PU53 10.10.10.0/24`
- Combined / disabled ARP: `nmap -sn -PE`

Input:

```
nmap -sn -PR 192.168.1.0/25 --packet-trace
-PE 10.10.X.0/24  nmap -sn -PE 10.10.X.100
nmap -sn -Pn 10.10.X.100
nmap -sn -PS22,80,443 10.10.X.0/24
nmap -sn -PA80,443 10.10.X.0/24
-PS -Pn -PU 10.10.20.0/24  nmap -sn
10.10.X.0/24 --disable-arp-ping
```

Target IP range (10.10.X.100 or 10.10.X.0/24)

Optional port list for -PS, -PA, -PU: 22, 80, 443

Output:

ARP

Nmap scan report for 10.10.X.1 host

0 up (10.0.0.20.1 (Gateway))

MAC Address: 02:08:75:5A:F2:XX

Nmap done: 256 IP addresses (2 hosts up)

ICMP

Nmap scan report for 10.10.X.1 host

0 up (0.00099s latency)

Nmap done: 1 IP address (1 host up)

Result:

The program was executed successfully.